

TITLE/INNOVATION NAME

NutriGene:An Integrated DNA Analysis and Recommendation System for Customized Nutrition Planning.

BACHELOR OF SCIENCE IN COMPUTER ENGINEERING

PREPARED BY:

BANAG, JOHN LESTER P.

INTRODUCTION

General dietary guidelines have been the foundation of public health for decades, advising populations on macro- and micronutrient intake to prevent chronic diseases such as type 2 diabetes, obesity, and cardiovascular issues (Gabriel et al., 2021). However, extensive research in nutritional science consistently demonstrates a high degree of **inter-individual variability** in metabolic responses to identical diets. A "**one-size-fits-all**" approach is increasingly recognized as insufficient, often leading to non-adherence, frustration, and clinically ineffective outcomes for significant portions of the population (Ghafouri et al., 2016; Swaminathan et al., 2023).

The field of **Nutrigenomics** has emerged as the scientific cornerstone for addressing this variability. It studies the interaction between an individual's genetic makeup (**genotype**) and their dietary components, revealing how specific gene variants influence nutrient absorption, metabolism, and disease susceptibility. This personalized approach, often called **Precision Nutrition**, requires sophisticated analysis of genetic markers, yet the current processes for translating complex genomic data into actionable, everyday dietary advice are often time-consuming, inaccessible, and dependent on highly specialized laboratory equipment (Gabriel et al., 2021; Swaminathan et al., 2023).

In rapidly evolving consumer markets, the demand for **immediate, accessible, and highly customized health data** is critical. Traditional genetic sequencing for nutrition typically involves sending a sample to a centralized lab, incurring high costs, and requiring a turnaround time of several weeks. This lengthy process disrupts the immediate feedback loop necessary for behavioral change and effective meal planning, making it impractical for point-of-care or direct-to-consumer (DTC) use in daily life. Therefore, the engineering challenge is to miniaturize and automate the entire genomic analysis-to-recommendation pipeline into a consumer-friendly, portable device (Raju et al., 2022).

Concurrently, advances in **portable molecular diagnostics** (such as smartphone-integrated DNA analysis and microfluidic lab-on-a-chip technologies) and **Artificial Intelligence (AI)** provide the necessary technical foundation. Deep learning models, particularly those in **Machine Learning (ML)**, excel at processing massive, complex datasets—like gene variant information paired with nutritional requirements—to generate highly specific, optimized meal plans. Combining rapid, on-site DNA sampling with a powerful AI processing unit represents a novel engineering opportunity: a handheld analyzer that bypasses the long wait times and high costs of traditional labs, generating immediate, genotype-based dietary recommendations (Anisuzzaman et al., 2022; Scebbba et al., 2022).

There is established work demonstrating the components of a comprehensive nutrigenomics device.

Existing Solutions in Genomics: The feasibility of miniaturized DNA analysis is proven by platforms such as the **MinION portable nanopore sequencer** (Oxford Nanopore Technologies), which can perform real-time sequencing and connects to a host device via USB-C. This technology validates the core concept of portable, fast, and cost-effective genetic reading (Raju et al., 2022). However, these devices are designed to output raw genomic data (sequencing information) and lack the direct integration of a consumer-facing application or the complex algorithms needed to translate that raw data into a structured, personalized meal plan.

This shows the feasibility of the miniaturized *hardware* but highlights a different use-case (research/diagnostics vs. consumer dietary guidance).

Existing Solutions in Personalized Nutrition (Non-Genomic): The market is saturated with digital platforms that offer personalized dietary recommendations using AI and ML (e.g., MyFitnessPal, diet tracking apps). These systems, often using **Collaborative Filtering** or **Content-Based Filtering**, show that data-driven recipe and meal planning algorithms are effective in promoting behavioral change (Kirsner et al., 2021; Rose et al., 2024). However, their input data relies primarily on **phenotypic** information (self-reported weight, activity, and dietary logs) rather than the foundational **genomic** markers. This approach is limited by the inherent biases and inaccuracies of self-reporting, underscoring the necessity of integrating an objective genetic input.

Existing Solutions in Clinical Genomics: Highly accurate and comprehensive nutrigenetic analysis is routinely performed by clinical laboratories using expensive, high-throughput sequencing machines (Schoeb et al., 2018; Filko et al., 2025). These systems provide the scientific backbone for nutrigenomics by identifying specific Single Nucleotide Polymorphisms (SNPs) associated with nutrient responses. While they prove the validity of the analysis, these designs remain **bulky, costly, and completely unsuitable for point-of-care** or at-home use. The clinical literature therefore provides the high-fidelity algorithms and scientific protocols useful for NutriGene, but does not directly provide a small-device blueprint.

Enabling Technology: Machine Learning for Nutrigenomics: The viability of connecting complex omics data (including genomics) to dietary outputs is strongly supported by recent literature on **Machine Learning (ML) in Precision Nutrition** (Anisuzzaman et al., 2022; Scebbba et al., 2022). ML models are increasingly used to cluster individuals by genetic risk, predict their response to specific nutrients, and even decompose compound ingredients in recipes to match a genetic profile. On-device models and edge accelerators make real-time inference on small hardware feasible, establishing the necessary computational backbone for an intelligent portable device.

Although nutrigenomic principles are well-established in research and the hardware for portable DNA analysis is commercially available (e.g., MinION), **there remains a critical gap in developing a single, integrated, lightweight, and affordable system** that performs rapid, point-of-care DNA extraction and directly translates the result into an actionable, AI-generated meal plan. Existing DTC companies offer genetic reports without the immediate, on-site DNA reading, while existing portable sequencers require complex data processing. This limitation underscores the need for an accessible, smart portable device, **NutriGene**, that bridges the gap between manual, generalized advice and expensive, slow, laboratory-based precision medicine.

Statement of the Problem (SOP)

Specifically, the study seeks to address the following questions:

1. How can a portable, saliva-based device be designed to perform **rapid DNA sampling and genetic detection** of key nutrigenomic Single Nucleotide Polymorphisms (SNPs)?
2. How can a **Machine Learning (ML)** algorithm be developed and optimized to accurately translate the detected genetic profile into a set of **personalized macronutrient ratios and dietary restrictions**?
3. How can the device be engineered to integrate the **portable molecular analysis system** (e.g., microfluidics) with the **on-device computing unit** (e.g., edge AI processor) to ensure a fast, real-time result?
4. How can the system deliver the personalized dietary output to the user in a **user-friendly, actionable format**, such as immediate recipe and meal recommendations?
5. How effective, precise, and user-friendly is the **NutriGene** prototype when tested in terms of its SNP detection accuracy, recommendation compliance with genetic data, and overall speed and usability?

Objectives of the Study

Main Objective

Therefore, this study aims to **design, develop, and test** the **NutriGene – DNA-Based Nutrition Analyzer: a Smart Portable Nutrigenomic System** capable of performing on-site genetic analysis and generating highly personalized, real-time dietary and meal recommendations. The system seeks to bridge the gap between expensive, time-consuming laboratory testing and immediate consumer-level precision nutrition.

Specific Objectives

To support this main objective, the following specific objectives are set to be addressed:

- **Design and Construct** a compact, handheld prototype integrating a **DNA detection module** (e.g., microfluidic cartridge and optical/electrochemical sensor) and a **low-power microcontroller** with an AI accelerator for portable, on-site analysis.
- **Develop** a **Genetic Detection Protocol** for key nutrigenomic SNPs (e.g., those related to caffeine metabolism, lactose tolerance, or saturated fat response) that can be reliably performed using the designed microfluidic platform.
- **Implement** a **Machine Learning (ML) Recommendation Engine** trained on a database of gene-nutrient interactions to automatically calculate personalized dietary targets (e.g., carbohydrate, protein, and fat percentages) based on the user's analyzed genetic profile.
- **Integrate** the hardware detection module with the software AI engine to create a seamless, end-to-end system that processes the raw genetic signal and outputs **immediate, actionable recipe suggestions** in under five minutes.
- **Conduct Performance and Usability Testing** to assess the device's **SNP detection accuracy**, the **compliance** of its recommendations with established nutrigenomic science, the **speed** of the analysis process, and the overall **user experience** and safety in a consumer setting.

This study focuses on the design, development, and testing of "**NutriGene**," an integrated **Smart Portable Nutrigenomic System** intended to provide **real-time, DNA-based personalized dietary recommendations**. The device is designed to perform two integrated functions: rapidly capture and analyze genetic markers from a non-invasive sample (like saliva) using portable molecular diagnostics, and subsequently translate that genetic profile into actionable meal guidance using an optimized Machine Learning algorithm. The scope includes the engineering of a **compact, handheld prototype** integrating a specialized DNA detection module (e.g., microfluidic sensor), a low-power microcontroller, and an on-device AI processor, all contained within a battery-powered chassis suitable for home or field use. The research aims to demonstrate the feasibility of combining genetic analysis and algorithmic recommendation in a single, accessible system, thereby bypassing the traditional costs and delays of centralized laboratory testing.

The study is delimited to a **specific, pre-defined panel of high-impact nutrigenomic SNPs** (Single Nucleotide Polymorphisms) relevant to common nutrient responses; **full-genome sequencing or the diagnosis of complex genetic diseases are outside the scope** of this research. The device is intended as a **personalized dietary guidance tool for general wellness** rather than a medical diagnostic or therapeutic instrument, meaning it does not replace the consultation of registered dietitians or professional medical care. Testing and evaluation will be conducted through **simulated sample analysis** using controlled synthetic DNA samples to assess detection accuracy, and through validated datasets to assess the precision of the **ML recommendation engine**, rather than long-term human clinical trials. Additionally, the study is limited to a prototype implementation utilizing commercially available components and **does not cover mass production, long-term durability testing**, or regulatory clearance for mass-market medical devices.

The development of **NutriGene** holds practical and societal importance, addressing the critical **inaccessibility gap** in precision health technology that currently limits personalized nutrition to expensive, slow laboratory services. For consumers, the device provides a safe, user-friendly solution that empowers individuals to manage their wellness proactively by making **genetically informed food choices**, which can lead to better dietary adherence and improved health outcomes compared to generalized diets. This accessibility can help in the prevention and better management of non-communicable diseases and contributes significantly to the **democratization of advanced health insights**. Additionally, the device can serve as a valuable educational tool in academic settings, offering students and researchers hands-on insight into the applied fields of **Nutrigenomics and Point-of-Care Engineering**.

From a technological perspective, this study contributes to the advancement of **Point-of-Care (PoC) systems and Edge Computing** by demonstrating the feasibility of integrating a low-power genetic sensor with a computationally intensive AI model in a compact, portable form factor. It successfully engineers a seamless **sensor-to-action workflow**, transforming raw molecular data into actionable dietary outputs in real time, which is a major innovation over current fragmented systems. This integration paves the way for further developments in autonomous, personal health-monitoring devices.

The study also aligns with key Sustainable Development Goals (SDGs). It supports **SDG 3: Good Health and Well-Being** by promoting timely, precise, and preventive health interventions through tailored nutrition. It advances **SDG 9: Industry, Innovation, and Infrastructure** through the creation of an innovative, integrated AI-powered health technology. Furthermore, it contributes to **SDG 10: Reduced Inequalities** by providing an

affordable and user-friendly system that expands access to high-quality personalized health information to underserved populations, fostering more equitable access to basic nutritional guidance. Overall, this study is expected to enhance the efficacy of personalized nutrition, improve health accessibility, and foster technological innovation in portable health systems.

Conceptual Framework

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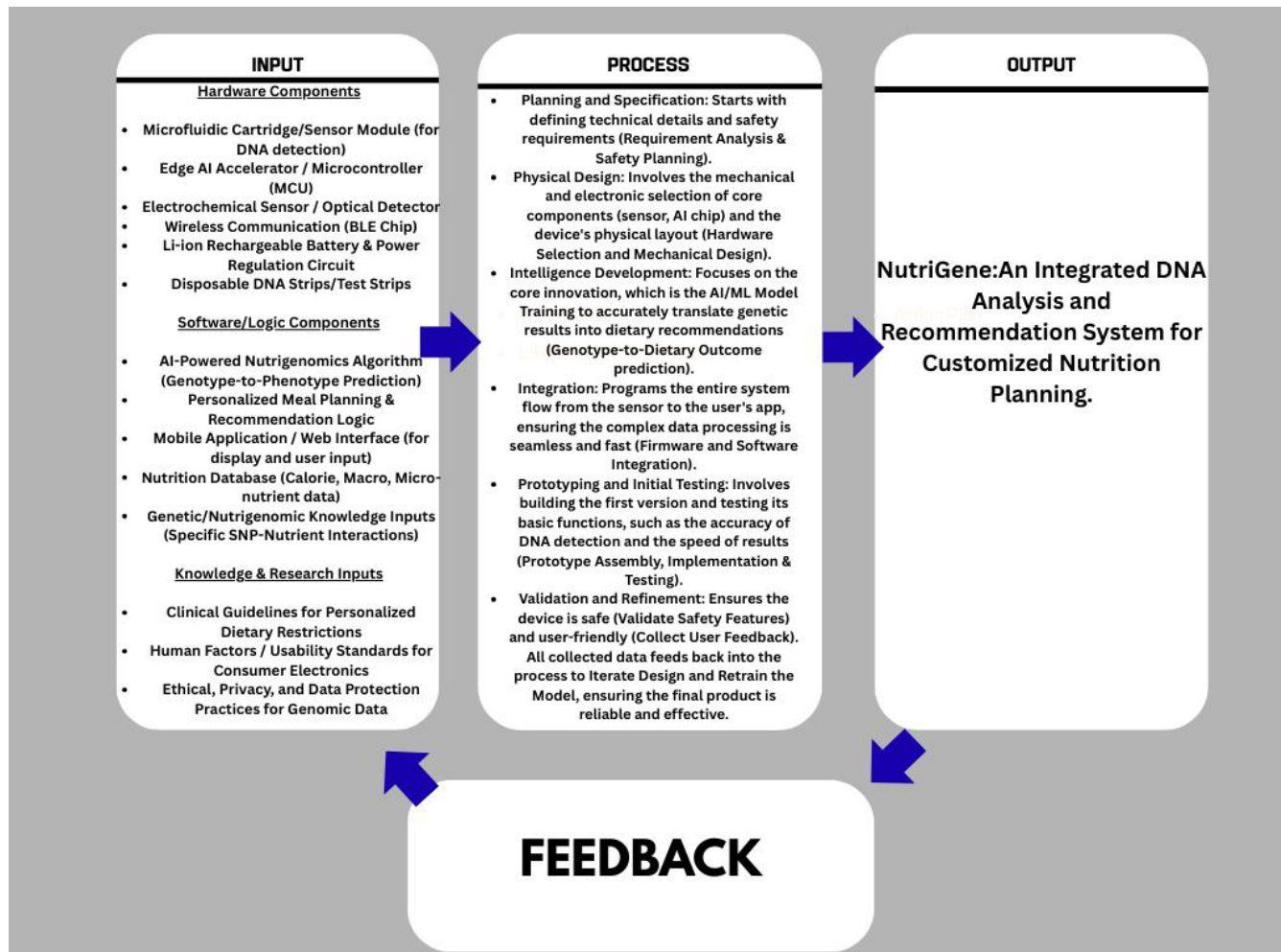


Figure 1. Conceptual Framework

The conceptual framework of **NutriGene – DNA-Based Nutrition Analyzer** shows how the system is developed from input to output.

The input includes the main **hardware components** like the sensor module, a microcontroller (MCU) or Edge AI accelerator, the wireless communication chip (BLE), the Li-ion rechargeable battery, and the core detection mechanism, such as an electrochemical sensor or microfluidic cartridge. It also includes **software and knowledge parts** such as the AI-powered nutrigenomics algorithm, which translates genetic data into nutrient requirements, the personalized meal planning logic, the nutrition database, and ethical/privacy standards for handling genomic data.

The **process** involves designing, developing, and testing the device from integrating the hardware and software, which includes training the **AI/ML model** on genotype-to-dietary outcome predictions. Key steps include assembling the handheld, thermometer-like prototype, integrating the sensor data with the on-device AI for rapid processing, and conducting performance tests to measure the accuracy of SNP detection and the **speed of analysis**. The process ensures that the system can reliably perform instant, on-the-go analysis without external lab delays.

The **output** is the **NutriGene – DNA-Based Nutrition Analyzer**, a portable device that instantly scans a saliva sample via disposable DNA strips and generates a genetically tailored, personalized daily meal plan. **Feedback** from accuracy testing, user experience, and safety validation is continuously collected and used to improve the system's design, optimize the AI model, and ensure high reliability and user safety.

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